

Yaohui Shu

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SUMMARY

Applied mathematics PhD candidate (M.S. Finance, CQF) working on stochastic processes, Monte Carlo simulation, and statistical inference. Derives closed-form SDE results and tests them against large simulations; first-author paper in *Geophysical Research Letters* (2026). Current methods work: causal discovery for nonstationary time series with feedback.

EDUCATION

Stony Brook University, Ph.D., Applied Mathematics & Statistics (GPA 3.78 / 4.0) Expected May 2027
Dissertation: *Uncovering Causal Mechanisms of Aerosol-Cloud-Turbulence Interactions Using Physics- and Data-Driven Methods*
Coursework: Advanced Stochastic Models & Risk Analysis, Numerical Analysis I–III, Financial Derivatives & Stochastic Calculus
University of Rochester, M.S., Finance (GPA 3.81 / 4.0) Aug 2021 – Aug 2022
Coursework: Derivatives Pricing, Financial Econometrics, Risk Management, Stochastic Calculus for Finance
CQF Institute, Certificate in Quantitative Finance (CQF)

SKILLS

Programming	Python (NumPy, SciPy, pandas, TensorFlow), MATLAB, C++, R; Git, SLURM/HPC
Simulation	Explicit and implicit SDE integrators, Monte Carlo with time-step and sample-size convergence testing, constrained boundary handling, large parameter sweeps on HPC
Stochastic Modeling	Stochastic calculus, Fokker-Planck and Langevin equations, Ornstein-Uhlenbeck processes, maximum entropy, Weibull and gamma distribution families
Statistics & ML	Causal discovery (FCI, PCMC, NOTEARS), kNN conditional mutual information, transfer entropy, doubly robust estimation (DML, AIPW, TMLE), LSTM, XGBoost, PCA
Finance	Derivatives pricing, financial econometrics, risk management, equity signal research, backtesting

RESEARCH EXPERIENCE

Stony Brook University Stony Brook, NY
Doctoral Researcher: Stochastic Modeling, Monte Carlo & Causal Inference Aug 2023 – Present

- Reconciled maximum-entropy theory with Langevin SDE dynamics for droplet size distributions: showed that their conflicting Weibull predictions for the same conservation constraint come from the choice of state variable.
- Generalized the SDE model to any conserved moment order under one hypothesis, reduced three equivalent formulations to a single system, and separated the moment that sets distribution shape from the one that sets scale.
- Validated the theory with Monte Carlo runs for orders 1 to 6 (10^5 paths each); traced the residual bias to the boundary rule and showed no rule can conserve probability, leave the driver untouched, and avoid mass at zero.
- Built a stochastic entrainment-mixing simulator with two limiting pathways from a common steady state, and a scheme that moves continuously between them by jointly controlling droplet number, mass, and mean volume.
- Designing causal discovery for time series with feedback and unobserved drivers: partial ancestral graphs under sigma-separation, nearest-neighbor independence tests, and surrogate-calibrated nonlinearity tests.
- Proposed a time-resolved causal-strength measure, the relative entropy rate between path measures with one drift dependence removed (closed form by Girsanov's theorem), meeting five of six stated postulates.

Teaching Assistant, AMS 595 Fundamentals of Computing (MATLAB, Python, C++) Aug 2026 – Present

Brookhaven National Laboratory Upton, NY
Research Assistant: Stochastic Processes, Monte Carlo & Information Theory Jan 2024 – Jan 2026

- Generalized a white-noise drift-diffusion model to a coupled SDE driven by an Ornstein-Uhlenbeck process spanning short to long memory, and showed that it converges to the white-noise model at fast decorrelation.
- Derived time-resolved information flow in closed form and built an ensemble estimator over 10^5 paths that matches theory, returns zero in the non-causal direction, and recovers direction where correlation cannot.
- Ran SLURM sweeps (10^5 paths, 5 seeds, 100 decorrelation rates, 20 noise levels) with conserving boundary handling, plus time-step and sample-size convergence studies; published in GRL with code on Zenodo.

CQF Institute Remote
Quantitative Finance Capstone: Equity Return Prediction 2023

- Built an LSTM direction model from 156+ momentum, volatility, and technical features, cut to 10–15 via XGBoost importance, self-organizing maps, and PCA; leakage-free splits and a 162-configuration search.

PUBLICATIONS & MANUSCRIPTS

- **Shu, Y.**, Liu, Y., Zhang, T., McGraw, R., Li, X. (2026). Stochastic Condensation with Information Flow for Causality Analysis. *Geophysical Research Letters*, 53(17), e2026GL123056. doi:10.1029/2026GL123056
- **Shu, Y.**, Liu, Y., Li, X. A Langevin-Type Stochastic Condensation Model to Explain the Size Distribution based on Maximum Entropy Principle. In preparation for *Physical Review E*; two more in preparation.
- Conference abstracts: AGU Fall Meeting 2025 (A21B-2034, first author; A21B-203); 106th AMS Annual Meeting 2026.